PROJECT REPORT

AIRLINE CUSTOMER SATISFACTION & SERVICE ANALYTICS

A Data Analytics Project

Prepared by: Mohammed Dhaqib MH
B.Sc. Statistics
Loyola College, Chennai
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Abstract

Customer satisfaction is an important factor in the airline industry because it influences passenger experience, loyalty and perceptions of service quality. This project analyses a public airline passenger satisfaction dataset containing 103,904 records and 25 columns. Microsoft Excel was used for initial data inspection, PostgreSQL and SQL for data storage and querying, Python for data processing and visualisation, and Power BI for interactive dashboard development. The analysis examines satisfaction across travel class, customer type, travel purpose, age groups, service ratings and departure-delay categories. It also compares service ratings between satisfied and neutral or dissatisfied passengers.

1. Introduction

Airlines operate in a competitive service environment in which passenger experience is shaped by booking, check-in, boarding, in-flight services, baggage handling and punctuality. Understanding how passengers evaluate these services can help identify strengths and opportunities for improvement.

This project applies a structured analytics workflow to a publicly available airline passenger satisfaction dataset. It combines data inspection, database management, SQL analysis, Python-based exploration and Power BI visualisation.

2. Project Objectives

- Inspect and understand a real-world airline passenger dataset.
- Identify missing values and duplicate records.
- Store and manage passenger data using PostgreSQL.
- Use SQL to calculate passenger counts, satisfaction rates and service statistics.
- Use Python and Pandas for data processing and visualisation.
- Compare satisfaction across passenger and travel segments.
- Examine service ratings and flight delays in relation to satisfaction.
- Develop an interactive Power BI dashboard and derive practical recommendations.

3. Dataset Description

The project uses a publicly available airline passenger satisfaction dataset obtained from Kaggle. The training file, train.csv, contains 103,904 passenger records and 25 columns. Variables include passenger demographics, customer type, travel type, class, flight distance, delay information, service ratings and satisfaction.

Attribute	Description

Records analysed	103,904
Number of columns	25
Data format	Structured CSV data
Target variable	satisfaction
Categories	satisfied; neutral or dissatisfied

This is a public sample dataset and does not represent United Airlines' confidential or internal customer data.

4. Tools and Technologies

Tool	Purpose
Microsoft Excel	Initial CSV inspection and data-quality checks.
PostgreSQL / pgAdmin	Database and passengers table creation.
SQL	Aggregation, filtering, grouping, averages and service-gap analysis.
Python	Data loading, processing, analysis and charts.
Pandas	Tabular data manipulation and exploratory analysis.
Matplotlib	Visualisation of selected results.
SQLAlchemy / psycopg2	Python-to-PostgreSQL connection and data transfer.
Power BI Desktop	Interactive dashboard, DAX measures and slicers.

5. Methodology

5.1 Data collection and inspection

The train.csv file was downloaded and opened in Excel. The structure and column names were reviewed, missing values were checked, and duplicate passenger IDs were examined.

5.2 Database creation and import

A PostgreSQL database named airline_analysis and a table named passengers were created. After the initial pgAdmin CSV import attempt was unsuccessful, a Python script read the CSV with Pandas, aligned column names with the database table, connected through SQLAlchemy and inserted records in batches. The import was verified using a SQL row count.

5.3 SQL analysis

SQL queries were used to count records, inspect missing arrival-delay values, group passengers by satisfaction, calculate satisfaction rates across categories, compare service ratings and examine average delays by satisfaction group. A common table expression was used for selected service rating-gap comparisons.

5.4 Python analysis

Pandas was used for dataset dimensions, missing-value analysis, duplicate checks and exploratory analysis. Matplotlib was used to create charts for selected comparisons.

5.5 Power BI dashboard

Power BI Desktop was connected to PostgreSQL using the local server and airline_analysis database. The passengers table was loaded, DAX measures were created for passenger counts and satisfaction percentages, and an Age Group calculated column was added. KPI cards, charts and slicers were arranged into a single-page interactive dashboard.

6. Data Quality Assessment

Check	Result
Total passenger records	103,904
Total columns	25
Missing Arrival Delay values	310
Duplicate passenger IDs found	0
Exact duplicate rows found	0

The 310 missing values were present in Arrival Delay in Minutes. They were not automatically replaced with zero because a missing delay value should not be interpreted as an on-time arrival. SQL AVG calculations exclude NULL values, so average arrival-delay calculations use available observations.

7. Analysis and Findings

7.1 Overall passenger satisfaction

Category	Passenger count	Percentage
Satisfied	58,879	56.67%
Neutral or dissatisfied	45,025	43.33%
Total	103,904	100.00%

The satisfied group represents 56.67% of the records, while 43.33% are neutral or dissatisfied. The latter proportion is substantial enough to merit further investigation.

7.2 Satisfaction by travel class

Travel class	Satisfaction rate
Business	69.43%
Eco	24.61%
Eco Plus	18.61%

Business Class recorded the highest satisfaction rate among the three travel classes.

7.3 Satisfaction by customer type

Customer type	Satisfaction rate
Loyal Customer	47.73%
Disloyal Customer	23.67%

Loyal customers recorded a higher satisfaction rate than disloyal customers.

7.4 Satisfaction by departure-delay category

Departure delays were grouped as On Time (0 minutes), Short Delay (1–15 minutes), Moderate Delay (16–60 minutes) and Long Delay (more than 60 minutes).

Delay category	Passengers	Satisfied passengers	Rate
On Time	58,668	26,842	45.75%
Short Delay	22,184	9,685	43.66%
Moderate Delay	15,813	5,911	37.38%
Long Delay	7,239	2,587	35.74%

The satisfaction rate was highest in the On Time category and declined across longer delay categories. This is an association and does not by itself establish causation.

7.5 Average service ratings

Service	Average rating
In-flight Wi-Fi	2.73
Online booking	2.76
Food and drink	3.20
Online boarding	3.25
Check-in service	3.30
Cleanliness	3.29
Leg-room service	3.35
In-flight entertainment	3.36
On-board service	3.38
Seat comfort	3.44
Baggage handling	3.63

Baggage handling recorded the highest average rating among the listed services, while in-flight Wi-Fi recorded the lowest.

7.6 Service rating comparison by satisfaction

Service	Neutral / dissatisfied	Satisfied	Gap
Online boarding	2.66	4.03	1.37
In-flight entertainment	2.89	3.96	1.07
In-flight Wi-Fi	2.40	3.16	0.76

Online boarding showed the largest rating gap among the three selected services, followed by in-flight entertainment and in-flight Wi-Fi. These gaps identify areas for investigation; they do not prove that service ratings cause satisfaction outcomes.

7.7 Average delays by satisfaction

Category	Average departure delay	Average arrival delay
Neutral or dissatisfied	16.50 min	17.13 min
Satisfied	12.61 min	12.63 min

Neutral or dissatisfied passengers recorded higher average departure and arrival delays. The arrival-delay average excludes records with missing arrival-delay values.

8. Power BI Dashboard

A single-page Power BI dashboard titled Airline Customer Satisfaction & Service Analytics was created. It combines high-level KPIs, segment comparisons, service ratings, delay analysis and interactive filters.

- KPI cards for total, satisfied, and neutral or dissatisfied passengers.
- Class satisfaction percentage.
- Customer satisfaction by customer type.
- Travel type satisfaction percentage.
- Customer satisfaction rate by age group.
- Satisfied passenger counts by travel type and customer type.
- Passenger satisfaction by departure-delay category.
- Average arrival and departure delays.
- Average passenger service ratings and selected service rating gaps.
- Slicers for satisfaction, customer type, class and type of travel.

DAX measures were created for total passengers, satisfied passengers, neutral or dissatisfied passengers and satisfaction percentage. An Age Group calculated column was created. Slicer interactions were tested so users could filter the report by passenger category.

9. Business Insights and Recommendations

9.1 Review the online boarding experience

Online boarding showed the largest rating gap between the satisfaction groups. Investigate usability, process clarity, check-in completion and boarding communication.

9.2 Examine in-flight entertainment

The rating gap suggests that content availability, system usability and technical reliability may be useful areas for further passenger-feedback analysis.

9.3 Investigate in-flight Wi-Fi

In-flight Wi-Fi had the lowest overall average rating among the services analysed. Connectivity reliability and passenger expectations could be examined.

9.4 Monitor punctuality and delay communication

Satisfaction rates were lower in longer departure-delay categories. Reviewing delay drivers and improving passenger communication may support a better experience.

9.5 Understand disloyal customers

The lower satisfaction rate among disloyal customers supports further investigation of their service experiences and feedback.

9.6 Maintain strong service performance

Baggage handling had the highest overall average rating. Maintaining performance in stronger service areas is important alongside improvement initiatives.

10. Limitations

- The project uses a public sample dataset, not internal airline data.
- The 310 missing arrival-delay observations may affect arrival-delay summaries.
- The analysis identifies descriptive patterns and associations, not causal effects.
- The dataset may not capture every factor that influences passenger satisfaction.
- Findings should not be generalised to all airlines or passenger populations without further validation.

11. Conclusion

This project demonstrates an introductory data analytics workflow applied to airline passenger satisfaction. The analysis found that 56.67% of passengers were satisfied and 43.33% were neutral or dissatisfied. Satisfaction differed across travel classes and customer types. Longer departure-delay categories were associated with lower satisfaction rates, while online boarding, in-flight entertainment and in-flight Wi-Fi showed notable rating gaps between satisfaction groups.

The work combined Excel, PostgreSQL, SQL, Python and Power BI to inspect, store, analyse and communicate data. The final dashboard provides an interactive view of the findings and supports further exploration of passenger segments and service areas.

12. Skills Demonstrated

- Data inspection and quality assessment.
- Exploratory data analysis and statistical interpretation.
- SQL querying, aggregation and grouping.
- PostgreSQL database creation and management.
- Python programming and Pandas data manipulation.
- Matplotlib data visualisation.
- Power BI dashboard development.
- DAX measures and calculated columns.
- Business insight generation and analytical communication.

13. Project Deliverables

- Original public airline passenger satisfaction CSV dataset.
- PostgreSQL database airline_analysis and passengers table.
- Python CSV-to-PostgreSQL import script.
- SQL analysis queries.
- Python analysis and visualisations.
- Power BI report: Airline_Customer_Satisfaction_Analysis.pbix.
- This project report.

Appendix A. Selected DAX Measures

```
Total Passengers =
COUNT('public passengers'[id])

Satisfied Passengers =
CALCULATE(
    COUNT('public passengers'[id]),
    'public passengers'[satisfaction] = "satisfied"
)

Neutral or Dissatisfied Passengers =
CALCULATE(
    COUNT('public passengers'[id]),
    'public passengers'[satisfaction] = "neutral or dissatisfied"
)

Satisfaction Percentage =
DIVIDE([Satisfied Passengers], [Total Passengers], 0) * 100
```

Appendix B. Reproducibility Note

This report summarises the workflow and results recorded during project development. Before formal academic submission, rerun the final SQL queries and compare their outputs with the tables in this report. Add the Kaggle dataset URL, access date, your name and any institution-specific formatting details required by your department.